

Kyrellos S. Ibrahim

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EDUCATION

Belmont University

B.S. in Computer Science, Minor in Business Systems & Analytics; GPA: 3.8/4.0

Nashville, TN

Aug 2022 – May 2026

EXPERIENCE

Application Engineer Intern

HCA Healthcare

April 2025 – Present

Nashville, TN

- **Modernized a legacy API used by multiple teams.** Rebuilt the service using FastAPI, SQLAlchemy, and Pydantic, reducing onboarding time for new developers and improving long-term maintainability.
- **Contributed to a cloud migration from Azure to GCP.** Leveraged Terraform to provision infrastructure as code, reducing technical debt and ensuring reproducible deployments.
- **Optimized complex SQL queries spanning millions of rows.** Designed performant queries to support analytics dashboards and business decision-making across the organization.
- **Built an end-to-end streaming data pipeline processing provider and facility updates every 10 minutes.** Ingested referral data from Pega via ADLS into Databricks, applied a multi-layer medallion architecture using SQL and PySpark, and structured gold-layer output as JSON payloads to update provider information on Google through a vendor API.

Undergraduate Research Assistant

Belmont University

May 2023 – July 2023

Nashville, TN

- **Simulated electromagnetic phenomena using Python.** Modeled gold nanospheres' interaction with light and validated results against Mie theory predictions, achieving general agreement between simulations and analytical solutions.
- **Contributed to a professor-led research team,** presenting findings and iterating on computational methodologies as part of a collaborative research effort.

PROJECTS

MunchFeed App | TypeScript, React Native, Firebase

Fall 2024

- **Built a full-stack social media app for sharing meals.** Developed core features including user authentication, image handling, and real-time database integration using Firebase.
- **Designed and implemented the UI across the entire app,** including comments, friends list, and profile views, ensuring a cohesive and intuitive user experience.
- **Executed thorough QA testing** to validate full app functionality and maintain UI consistency across devices.

ItemShopBot | GitHub | Python, Discord.py, Fortnite API

Spring 2025

- **Built a Discord bot that delivers daily Fortnite item shop data.** Integrated with the Fortnite API to automatically fetch and display current shop inventory to server members.
- **Implemented a per-user wishlist system with persistent storage,** allowing players to track desired items and receive notifications when they appear in the shop.
- **Architected the bot using Discord.py's cog pattern** with separated modules for shop logic, storage, and commands, following clean separation of concerns.

Game Updates vs. Player Sentiment | GitHub | Python, Jupyter, Pandas

Fall 2025

- **Conducted a data-driven sentiment analysis study.** Analyzed how controversial game updates affect player engagement across titles including Overwatch 2, RuneScape, and Helldivers.
- **Performed NLP sentiment analysis on Reddit and Twitter data** to quantify community mood shifts before, during, and after major update announcements.
- **Applied time series and comparative analysis** to test four competing hypotheses, identifying patterns in short-term outrage versus long-term player retention.

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, SQL, Java, Kotlin, C#, HTML/CSS

Frameworks & Libraries: React, React Native, FastAPI, PySpark, Firebase, SQLAlchemy, Pydantic, Discord.py, Tailwind CSS

Tools & Platforms: Git, GitHub, Docker, Terraform, Databricks, Bash, Unix/Linux, Jupyter Notebook, SSMS, npm, Xcode